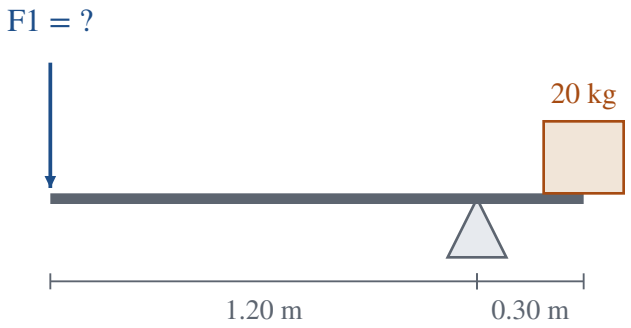
8th Grade Physical Science - Simple Machines

Name: _____ Date: _____

Ideal machines. Use g = 10 N/kg. Ignore friction and liquid height differences.

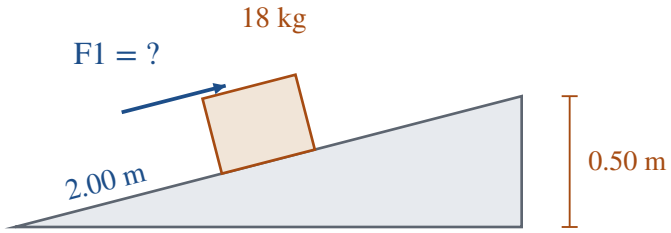
$F_1 d_1 = F_2 d_2$ $MA = F_{out} / F_{in} = d_{in} / d_{out}$

1. Lever



F_1 = _____ N MA = _____

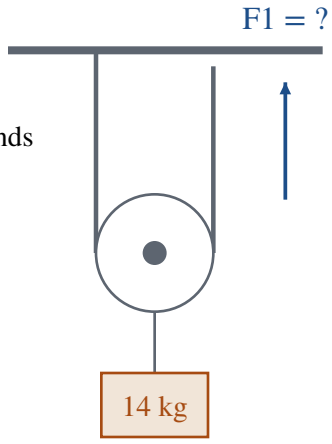
2. Ramp



F_1 = _____ N MA = _____

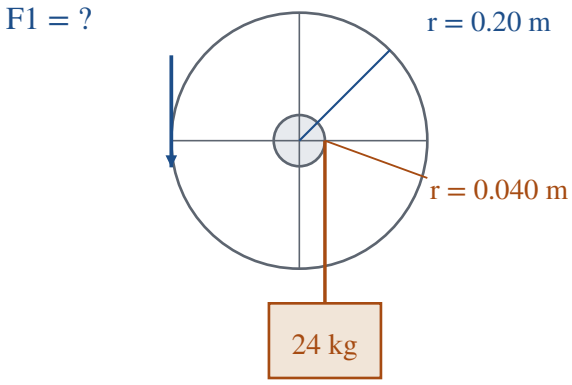
3. Movable pulley

2 supporting strands



F_1 = _____ N MA = _____

4. Wheel and axle



F_1 = _____ N MA = _____

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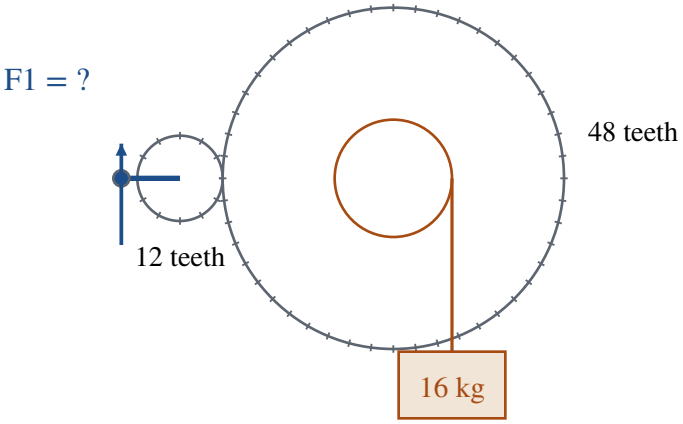
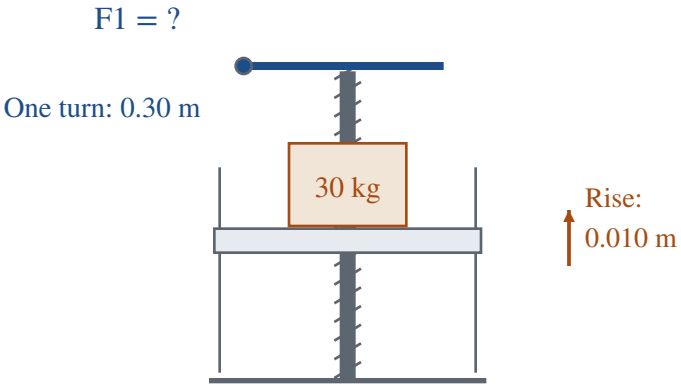
5. Geared winch



Equal crank and drum radii

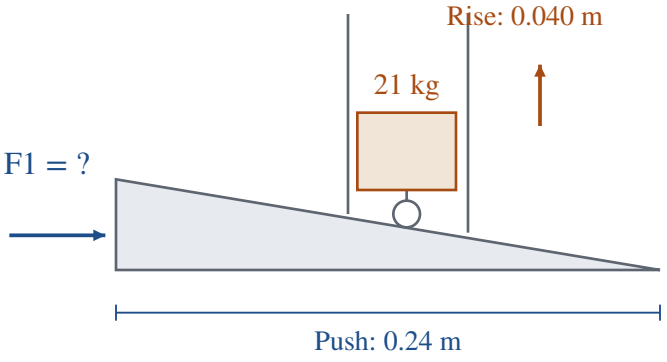
F_1 = _____ N $\quad$ MA = _____

6. Screw



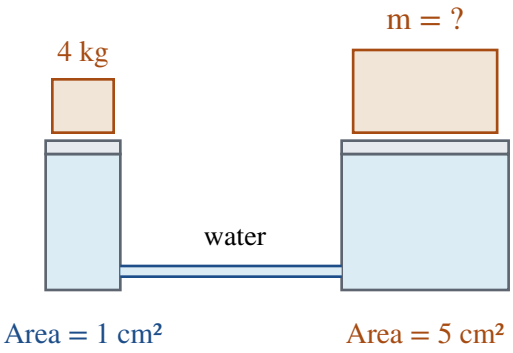
F_1 = _____ N $\quad$ MA = _____

7. Wedge



F_1 = _____ N $\quad$ MA = _____

8. Linked syringes



m = _____ kg $\quad$ MA = _____

Find the balanced output mass.